PROFESSIONAL TRAINING PROGRAM

🐍 Python for Hackers

Complete Master Curriculum & Advanced Offensive Automation Guide

TOTAL CURRICULUM	SKILL LEVEL	FOCUS AREA
16 Core Modules + 6 Projects	**Beginner to Advanced**	**Offensive Security & Automation**

COURSE DESCRIPTION & OBJECTIVE

This premium curriculum is engineered to transform developers, security enthusiasts, and cybersecurity professionals into proficient Python script writers capable of building custom penetration testing tools, network scanners, exploit payloads, automated log analyzers, and scalable security automation utilities.

CORE CURRICULUM BREAKDOWN

MODULE 01 — Python Fundamentals

- What is Python?
- Python interpreter & scripts
- Type casting
- Comments & Operators
- PEP 8 & clean coding guidelines
- Installing Python & VS Code
- Variables & Data types
- Input & output
- Basic debugging

MODULE 02 — Conditions & Loops

- `if`, `elif`, `else` statements
- `for` loops & iterations
- `break`, `continue`, `pass`
- Nested loop execution
- Nested conditional logic
- `while` loops
- Sequence generation with `range()`
- Practical algorithmic problem solving

MODULE 03 — Python Data Structures

- Lists & Tuples
- Strings manipulation
- List, dict & set comprehensions
- Selecting optimal data structures
- Sets & Dictionaries
- Indexing & Slicing techniques
- Nested data structures

MODULE 04 — Functions & Modular Design

- Creating & defining functions
- Parameters & return values
- Default arguments
- Variable arguments: `*args` and `**kwargs`
- Variable scope (Local vs Global)
- Lambda functions
- Recursive functions
- Modular programming best practices

MODULE 05 — Error Handling & Debugging

- Syntax vs Runtime vs Logical errors
- `try` / `except` blocks
- `else` & `finally` clauses
- Raising exceptions with `raise`
- Creating custom Exception classes
- Advanced debugging & tracing techniques

MODULE 06 — Files & Data Handling

- File reading, writing & appending
- Directory manipulation
- Working with CSV & JSON files
- Parsing TXT & raw log files
- System interactions via `os` module
- Modern path operations with `pathlib`
- Building automated file processors

MODULE 07 — Object-Oriented Python

- Classes, Objects & Constructors
- Attributes & Methods
- Instance vs Class variables
- Inheritance & Method Overriding
- Encapsulation & Data Hiding
- Polymorphism
- Magic (Dunder) methods
- Building modular, reusable tools

MODULE 08 — Modules & Packages

- Python import system
- Exploring built-in modules
- Creating custom modules & packages
- Package management with `pip`
- Virtual environments (`venv`)
- Managing `requirements.txt`
- Dependency resolution

MODULE 09 — Command Line Architecture

- Parsing arguments with `sys.argv`
- Creating interactive CLI applications
- Configuring flags, switches & options
- Building production-grade security tools
- Professional CLI with `argparse`
- Building menu-driven tools
- Strict input validation & Exit codes

MODULE 10 — Regular Expressions (Regex)

- Regex fundamentals & syntax
- Capturing groups & assertions
- Search, match & replace operations
- Extracting structured security data
- Character classes & Quantifiers
- Pattern matching using `re` module
- Data validation patterns

MODULE 11 — Working With APIs

- REST API fundamentals
- GET, POST, PUT, DELETE methods
- Parsing JSON responses
- API Authentication (Tokens, API Keys, OAuth)
- HTTP requests via `requests` library
- Custom Headers & URL parameters
- Handling HTTP status codes & errors
- Building custom threat intelligence clients

MODULE 12 — Networking With Python

- IP addresses, Ports & Protocols
- TCP socket client & server creation
- Client/Server network architecture
- Building port scanners & banners grabbers
- Low-level networking with `socket`
- UDP socket fundamentals
- Socket timeouts & Exception handling

MODULE 13 — Multithreading & Concurrency

- Concurrency vs Parallelism
- Thread synchronization & Locks
- Multiprocessing fundamentals
- Selecting Threads vs Async for security tools
- Multithreading using `threading`
- Thread pools with `concurrent.futures`
- Asynchronous programming (`asyncio`)

MODULE 14

Subprocess & System Automation

- Executing commands via `subprocess`
- Environment variables manipulation
- Executing external tools safely
- Capturing STDOUT & STDERR output
- Automating OS administration tasks
- Building end-to-end security workflows

MODULE 15

Advanced Python Techniques

- Iterators & Custom Generators
- `map()`, `filter()`, `reduce()`
- Advanced structures in `collections`
- Type Hints & `dataclasses`
- Decorators & Context Managers
- `zip()` & `enumerate()`
- `itertools` for combination attack tools

MODULE 16

Professional Development

- Standard repository & project architecture
- Configuration files & `.env` management
- Documentation & docstrings
- Packaging & distribution
- Advanced logging framework implementation
- Unit testing using `pytest`
- Git / GitHub version control workflow
- Writing clean, maintainable codebases

🚀 PRACTICAL CAPSTONE PROJECTS

PROJECT 01 — Professional CLI Security Toolkit

Students construct a modular, menu-driven CLI application designed to aggregate various security utilities into a unified suite.

OOP Design **Argparse** **Logging** **Environment Config**

PROJECT 02 — Automated File Operations & Sanitizer Tool

Develop an automated tool that scans deep directory structures, categorizes sensitive file formats, renames artifacts, and produces structured analytical summaries.

Pathlib **OS Module** **Batch Processing** **Report Generator**

PROJECT 03 — Multi-Format Security Log Analyzer

Engineered to parse web server and firewall logs, extract suspicious IP/user-agent patterns using regex, calculate statistics, and export security reports.

Regex | JSON/CSV Handling | Pattern Matching | Statistics

PROJECT 04 — Threat Intelligence API Client

Build an asynchronous API client that queries threat intelligence databases (e.g., VirusTotal / AbuseIPDB), processes JSON responses, and manages rate limits gracefully.

Requests | Authentication | JSON Parsing | Error Recovery

PROJECT 05 — Multi-Threaded Network Client/Server Engine

Build a custom concurrent TCP socket client and multi-threaded server architecture supporting encrypted messaging, heartbeats, and multi-client routing.

Socket Programming | TCP/IP | Threading | Connection Management

PROJECT 06 — High-Performance Multithreaded Scanner

Construct a high-concurrency reconnaissance tool leveraging worker pools and thread queues for fast network probing and vulnerability evaluation.

Thread Pools | Queue System | Concurrent Futures | Performance Metrics